28/07/2025

Experiment No: 1

Aim: Develop the program to generate continuous waveforms & perform basic operations signals on MATLAB IDE.

TOOLS REQUIRED:

MATLAB IDE

MATLAB PROGRAM

1a) %. Generate a 50 Hz square wave using MATLAB function

```
clc;
clear all;
%. Enter the time vector
t = 0:0.001:0.625
x = square (2 *pi* 50 * t);
plot (t, x)
title ('Square wave pulse');
xlabel ('time');
ylabel ('Amplitude');
```

Output: After execution of the program graphical window displays 50Hz Analog Square wave.

d d m m y y y y

1b) % Generate saw tooth, exponential, sine & cosine waveforms employing MATLAB signal Processing commands

```
clc;
clear all;
fs = 120;
t = 0:1/fs:1;
y = sawtooth (2*pi* 40 * t);
subplot (2,2,1);
plot (t, y);
xlabel ('t');
ylabel ('y(t)');
title ('Sawtooth Signal');
t = 0:1:10;
y = 0.4.^(t);
 subplot (2,2,2);
plot (t,y);
xlabel ('t')
ylabel ('y(t)');
title ('Exponential Signal');
t = 0:0.01:pi;
y = sin (2*pi* t);
subplot (2,2,3);
plot(t,y);
xlabel ('t');
ylabel ('y(t)');
title ('Sine signal');
t = 0:0.01:pi;
y = cos(2*pi*t);
```

d d m m y y y y

```
subplot (2, 2, 4);
plot (t, y);
xlabel ('t');
ylabel ('y(t)');
title ('Cosine Signal');
```

Output : After execution of the program graphical window displays Sawtooth, exponential, Sine & Cosine waveforms.

4	d	0	2	2	0	2	5

Experiment No : 2

2a) Develop the Program to generate discrete waveforms & perform basic operations on signals on MATLAB IDE.

Tools Required :

MATLAB IDE.

MATLAB PROGRAM

```
%. Program to generate discrete wave forms - Impulse, step, ramp & discrete signal.
clc;
clear all;
t = -3 : 1 : 3;
y = [zeros(1,3), ones(1,1), zeros(1,3)];
subplot(2,2,1);
stem(t,y);
xlabel('t');
ylabel('y(t)');
title('Unit Impulse Signal');
n = input('Enter the n value');
t = 0 : 1 : n-1;
y = ones(1,n);
subplot(2,2,2);
step(t,y);
xlabel('t');
ylabel('y(t)');
title('Unit Step Signal');
y = [1 2 4 6 8];
subplot(2,2,3);
```

```
stem(y);
xlabel('t');
ylabel('y(t)');
title('Discrete Signal');
n = input('Enter the length of ramp sequence');
t = 0:n;
subplot(2,2,4);
stem('t');
xlabel('t');
ylabel('y(t)');
title('Ramp Signal');
```

OUTPUT : After execution of the program graphical window displays discrete wave forms - Impulse, step, ramp & discrete signal.

04 02 2024

2b) Consider any two sinusoidal input signals of your own & carry out the following basic operations of these two input signals:

1) Addition 2) Subtraction 3) Multiplication 4) Scaling

Tools Required: MATLAB IDE

MATLAB PROGRAM:

```
%. Program to perform the mathematical operation on
two sinusoidal signals.
clc;
clear all;

n = 0: 0.01: pi;
y = sin(2*pi*n) + sin(3*pi*n);
stem(n, y);
xlabel('n');
ylabel('y[n]');
title('Addition of two given input signals');
z = sin(2*pi*n) - sin(3*pi*n);
figure(2);
stem(n, z);
xlabel('n');
ylabel('z[n]');
title('Subtraction of two given input signals');
w = sin(2*pi*n). * sin(3*pi*n);
figure(3);
stem(n, w);
xlabel('n');
ylabel('w[n]');
```

d d m m y y y y

```
title ('Multiplication of two given input signals');
p = 5*(sin(2*pi*n) + sin(3*pi*n));
figure (4);
stem (n, p);
xlabel ('n');
ylabel ('p[n]');
title ('Scaling of addition of given two input signals');
```

OUTPUT: After execution of the program graphical window displays mathematical operations (Additions, Subtraction, Multiplication & scaling) on two sinusoidal signals.

28 Dec 2026

Experiment No : 3

Develop the program to determine linear convolution using MATLAB IDE.

Tools Required : MATLAB IDE.

MATLAB PROGRAM :

```
%. Program to perform the linear convolution of two
discrete time sequences.
clc;
clear all;
x = input ("Enter the value of x[n]');
h = input ('Enter the value of h[n]');
y = conv (x, h);
disp (y);
figure (1);
stem (y);
title ('Linear Convolution of two discrete sequences');
xlabel ('n');
ylabel ('y[n]');
```

OUTPUT : After execution of the program, graphical window & figure window displays the linear convolution of two discrete sequences

$x[n] = [1 \quad 2 \quad 3 \quad 4]$; $h[n] = [5 \quad 6 \quad 7 \quad 8]$;

$y[n] = x[n] * h[n] = \{5, 16, 34, 60, 61, 52, 32\}$

04 02 2026

Experiment No : 4

Develop the program to determine the circular convolution using MATLAB IDE.

Tools Required : MATLAB IDE

MATLAB PROGRAM :

```
% Program to perform the circular convolution of
two discrete time periodic sequences.
clc;
clear all;
x = input ('Enter the value of x[n]');
h = input ('Enter the value of h[n]');
N = input ('Enter the length of resulting vector');
y = cconv (x, h, N);
disp (y);
figure (1);
stem (y);
title ('Circular Convolution of Two discrete time
periodic sequences');
xlabel ('n');
ylabel ('y (n)');
```

Output : After execution of the program graphical window & figure window displays the circular convolution of two discrete time periodic sequences.

$x[n] = [1\ 2\ 3\ 4]$; $h[n] = [5\ 6\ 7\ 8]$;

$y[n] = x[n] \circledast h[n] = \{66, 68, 66, 60\}$

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Experiment No - 5

Develop the program to verify Sampling theorem (All three cases) using MATLAB IDE.

Tools Required : MATLAB IDE

MATLAB PROGRAM :

```
% Program to verification of the sampling theorem
for all three cases.
clc;
clear all;
tfinal = 0.05;
t = 0:0.00005:tfinal;
fd = input('Enter analog frequency');
% define analog signal for comparison
xt = sin(2*pi*fd*t);
% Simulate condition for undersampling i.e., fs 1<2*fd
fs1 = 1.3* fd;
% define the time vector
n1 = 0.1/fs1:tfinal;
% Generate the undersampled signal
xn = sin(2*p*n1*(fd);
% Plot the analog & sampled signals
subplot(3,1,1);
plot(t,xt,'b',n1,xn,'r*-');
title('Undersampling plot');
xlabel('time');
ylabel('amplitude');
legent('analog','discrete');
```

d d m m y y y y

```
% condition for Nyquist plot
fs2 = 2*fd;
n2 = 0:1/fs2:tfinal;
xn = sin(2*pi*fd*n2);
subplot(3,1,2);
plot(t, xt, 'b', n2, xn, 'r*-');
title('Nyquist plot');
xlabel('time');
ylabel('amplitude');
legend('analog','discrete');
% condition for oversampling
fs3 = 5*fd;
n3 = 0:1/fs3:tfinal;
xn = sin(2*pi*fd*n3);
subplot(3,1,3);
plot(t, xt, 'b', n3, xn, 'r*-')
title('Oversampling plot');
xlabel('time');
ylabel('amplitude');
legend('analog', 'discrete.
```

Input : Enter the analog frequency : 500Hz

Output : After execution of the program graphical window & figure window displays all three cases (1. Under sampling 2. Nyquist Sampling 3. Over Sampling) of sampling theorem.

2 5 0 2 2 0 2 5

Experiment No - 6

Develop the program for N point DFT of a given discrete time sequence & to plot magnitude & phase spectrum using MATLAB IDE.

Tools Required: MATLAB IDE.

```
clc;
clear all;
x = input ('Enter the discrete time sequence');
y = fft(x);
disp (y);
disp (M);
theta = angle (y);
disp (theta);
figure (1);
stem (M);
title ('Magnitude of Spectrum');
xlabel ('n');
ylabel ('Magnitude');
figure (2);
stem (theta);
title ('angle of spectrum');
xlabel ('n');
ylabel ('angle');
```

OUTPUT: After execution of the program graphical window & figure window displays the magnitude & phase of the spectrum. x[n] = [1 2 3 4] y = [10 -2+2i -2 -2-2i]

M = 10, 2.828, 2, 2.828 theta (In radians) = 0, 2.355, 3.14, -2.282

25/12/2023

Experiment No -7

Develop a program to determine linear convolution of two given discrete sequences using DFT & IDFT algorithms on MATLAB IDE.

Tools Required : MATLAB IDE

```
clc;
clear all;
L = input('Enter the number of input elements');
M = input('Enter the number of impulse elements');
N = L+M-1;
x = input('Enter the value of x[n]');
h = input('Enter the value of h[n]');
x1 = fft(x,N);
x2 = fft(h,N);
z = x1.*x2;
y = ifft(z);
disp(y);
```

OUTPUT : After execution of the program graphical window & figure window displays the linear convolution of two discrete time sequences using DFT & IDFT algorithm

$x[n] = [1\ 2\ 3\ 4]$; $h[n] = [5\ 6\ 7\ 8]$;

$y[n] = x[n] * h[n] = \{5, 16, 34, 60, 61, 52, 32\}$

2501 02 2026

Experiment No : 8

Develop a program to determine circular convolution of two given discrete time sequences using DFT & IDFT algorithms on MATCAB IDE.

Tools Required : MATCAB IDE

```
clc ;
clear all ;
x = input ('Enter the value of x[n]');
h= input ('Enter the value of h[n]');
N = input ('Enter the length of resulting vector');
y1 = fft (x, N);
y2 = fft (h, N);
y3 = y1.* y2;
y = ifft (y3);
disp (y)
figure (1);
stem (y);
title ('circular convolution using DFT & IDFT method');
xlabel ('n');
ylabel ('y[n]');
```

Output : After execution of the program graphical window & figure window displays the circular convolution of two discrete time sequences using DFT & IDFT algorithms.

x[n] = [1 2 3 4] ; h[n] = [5 6 7 8] ;

y[n] = x[n] ⊛ h[n] = {66, 68, 66, 60}